Adversarial Attacks on Autonomous Vehicles: Camera-Only and Cross-Attention Fusion Models Under Single-Sensor Attack

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I. Introduction

The fast growth of autonomous vehicles has created a new type of important AI system that depends on sensors like cameras and LiDAR to understand the world around it. Cameras captures visual details, while LiDAR adds 3D geometric depth. As these systems move toward higher levels of autonomy with less human supervision, the security of perception becomes a serious concern. A single mistake in perception, like misclassifying an object, missing an obstacle, or predicting the wrong driving action can have dangerous consequences.

One of the biggest threats to these systems is adversarial attacks. Unlike hardware failures or bad weather, adversarial attacks are intentional. The Fast Gradient Sign Method (FGSM), introduced by Goodfellow et al. (2014), showed that neural networks are highly vulnerable to small, mathematically crafted perturbations that are invisible to humans but cause the model to produce completely wrong predictions. In autonomous vehicles, this could mean the car runs a stop sign, misses a turn, or ignores an obstacle.

Multi-sensor fusion is often treated as the obvious way to improve perception reliability because it gives the model access to complementary sensor information. However, fusion does not automatically guarantee robustness. Prior work on camera–LiDAR fusion has shown that fusion models can be more accurate and more robust than single-sensor models under some single-source attacks, but they remain vulnerable and require careful adversarial evaluation (Wang et al., 2022). This means that clean LiDAR may not automatically protect a fusion model if the camera branch is adversarial corrupted.

To test which model is more reliable, this research uses a 2×2 design with four models: a standard camera-only model, an adversarially trained camera-only model, a standard cross-attention fusion model, and an adversarially trained cross-attention fusion model. In all cases the camera is attacked using FGSM while the LiDAR in the fusion models stays clean. The central research question is: how does adversarial training affect robustness for camera-only versus camera-LiDAR fusion when only the camera is under attack?

II. Motivation

The motivation for this research comes from a debate that has been going on in the AV industry: does adding another sensor actually improve security, or does the model still need to be trained to handle adversarial corruption? The 2018 Uber autonomous vehicle crash in Tempe, Arizona illustrates why sensor availability alone is not enough. The National Transportation Safety Board reported that the automated driving system detected the pedestrian 5.6 seconds before impact, but it did not accurately classify her as a pedestrian or predict her path in time to prevent the crash (NTSB, 2019). Having sensors does not guarantee the model will use their information correctly under difficult or conflicting conditions.

This question is also important because not all AV companies have the same resources. Tesla is a well-known example of a company that has pursued a camera-centered perception strategy. Tesla, for example, uses a camera-only perception strategy supported by massive fleet data, specialized hardware, and continuous deployment feedback (Tesla, n.d.). That works for them, but smaller companies or research teams building AV systems often cannot match that scale. For these groups, camera-LiDAR fusion is attractive because LiDAR provides 3D structure that complements camera features without requiring billions of training miles. The problem is that fusion only helps if the model actually knows how to use LiDAR when the camera becomes unreliable.

I think it is important to mention that Tesla has explicitly argued against LiDAR. Elon Musk called it a "crutch" and suggested that any company relying on LiDAR is "doomed" (Korosec, 2019). From a security perspective, that argument creates an interesting question: can adding more sensors create more failure points when the model must resolve conflicting information? If the system must solve conflicting signals from multiple sensors, it may perform worse than a simpler single-sensor system if the fusion mechanism has not been trained for those conflicts. The results in this project provide some evidence for that concern: under adversarial attack, the standard fusion model performed worse than the standard camera-only model. This does not mean LiDAR is harmful in general, but it does support the idea that adding more sensors without robustness-aware training can introduce new vulnerabilities rather than fixing existing ones.

Recent research has made this topic more important. Cheng et al. (2024) showed that camera–LiDAR fusion models can be compromised through the camera modality alone, even when the LiDAR input remains benign. Their work challenges the assumption that fusion automatically protects the model just because one sensor remains clean. This motivates the central concern of this project: if the camera becomes unreliable, the model must not only have access to LiDAR but must also know how to use it correctly under sensor conflict.

III. Related Work

Sensor fusion has been proposed to improve perception accuracy by combining information from different sensors. Under clean conditions, fusion models often outperform single-sensor models on standard benchmarks. Wang et al. (2022) found that camera–LiDAR fusion can improve both accuracy and robustness compared with single-sensor models in many scenarios, although fusion models still require adversarial evaluation because they remain vulnerable to targeted attacks.

Attention-based fusion has become a popular direction because it allows one modality to selectively query another instead of blindly combining features. Vaswani et al. (2017) introduced attention mechanisms to learn relationships between tokens, and later autonomous-driving fusion models such as TransFusion used attention-based camera–LiDAR fusion to improve detection performance under degraded conditions (Bai et al., 2022). These works motivate the cross-attention design used in this project.

Recent work has begun studying single-modality attacks against camera–LiDAR fusion. Cheng et al. (2024) showed that fusion models can be attacked through the camera modality alone, while Wang et al. (2022) found that fusion can improve robustness against some single-source attacks but still involves important training tradeoffs. In real-world camera attacks, the attacker usually does not access the camera software directly. Instead, they place a physical or printable adversarial patch inside the camera's field of view, such as on a vehicle, sign, or roadside object. The camera captures the scene normally, but the patch changes the image features seen by the model. My FGSM attack is not a physical patch attack; it is a simpler digital simulation of the same sensor-conflict idea, where the camera input is corrupted while LiDAR remains clean. My work builds on this direction by using a controlled 2×2 comparison: camera-only versus camera–LiDAR fusion, each with and without adversarial training. This allows the experiment to separate the effect of adding LiDAR from the effect of adversarial training.

IV. Methodology

This project uses the KITTI raw dataset to create a simplified autonomous vehicle classification task. KITTI provides synchronized camera, LiDAR, GPS, and IMU/OxTS data collected in real driving environments, making it suitable for studying both single-sensor and multi-sensor perception models (Geiger et al., 2013). Each frame gets one of four labels: straight, turn_left, turn_right, or stop. Labels come from the OxTS motion data: stop when forward velocity $v_f < 1.0$, turn_left when yaw rate $w_z > 0.03$, turn_right when $w_z < -0.03$, and straight otherwise. This simpler task lets the experiment focus on adversarial robustness and sensor-fusion behavior rather than full vehicle control.

A controlled stratified 80/20 frame-level split was used so all four classes appear in both training and testing. The final training set contains 948 straight, 166 turn_left, 128 turn_right, and 174 stop frames. The test set contains 237 straight, 41 turn_left, 32 turn_right, and 44 stop frames. The same data split, preprocessing, FGSM implementation, and evaluation metrics are used across all four models so the comparisons are fair.

A. Camare-Only Models

Both camare-only models use a ResNet18 backbone with ImageNet pretrained weights (He et al., 2016). The original classification layer is replaced with a custom head that predicts one of the four driving action labels. The standard model trains on clean images only. The adversarially trained model uses the same architecture and setup but modifies each training batch so 50% of examples are FGSM-attacked camera images while the other 50% remain clean. Both models train for 15 epochs using cross-entropy loss, AdamW optimization, cosine learning rate scheduling, and a square root weighted sampler to reduce class imbalance without over-amplifying minority classes. Training images use random cropping and color jitter; test images use only resize and normalization.

B. Cross-Attention Fusion Models

The fusion models combine a camera branch and a LiDAR branch using multi-head cross-attention. The camera branch is the same ResNet18 backbone, projecting to a 256-dimensional query vector. The LiDAR branch divides each point cloud into eight angular sectors around the vehicle (front, front-left, left, rear-left, rear, rear-right, right, and front-right). Each sector is sampled to a fixed number of points, normalized, and processed through a shared PointNet-style MLP (Qi et al., 2017) to produce eight 256-dimensional tokens. These sector tokens act as keys and values in the cross-attention layer, while the camera feature acts as the query. This makes the attention weights meaningful since each weight corresponds to an actual spatial region around the vehicle, which makes it possible to analyze how the model uses LiDAR under clean and adversarial conditions.

The standard fusion model trains on clean camera-LiDAR. The adversarially trained version uses the same architecture and training setup, but half of each batch has FGSM-attacked camera input while LiDAR stays clean. This directly mirrors the sensor-conflict scenario being studied. Both fusion models use AdamW optimization, cosine scheduling, cross-entropy loss, and balanced accuracy checkpointing to avoid selecting a model that just predicts the majority class.

C. Adversarial Attack Procedure

FGSM perturbs the input in the gradient sign direction to increase the model's loss (Goodfellow et al., 2014). For camera-only models, FGSM is applied to the camera image. For fusion models, FGSM is applied only to the camera while LiDAR stays clean; this is the sensor-conflict scenario. Both adversarially trained models use epsilon = 0.05 during training with a 50/50 adversarial ratio. Seven epsilon values (0.0, 0.01, 0.03, 0.05, 0.1, 0.2, 0.3) are used at evaluation time to show how performance changes as attack strength increases.

D. Final 2x2 Comparison

After the camera-only and fusion experiments are run separately, a third notebook combines results for the final comparison. No training happens there, because it just loads the saved results and compares all four models' side by side. The four conditions are: standard camera-only, adversarially trained camera-only, standard fusion, and adversarially trained fusion. This design makes it possible to isolate the effect of LiDAR fusion from the effect of adversarial training, and to see whether combining both gives additional benefit.

The notebooks also generated additional plots, prediction distributions, visual attack examples, and saved output files. Since the report is intended to summarize the main findings, only the most relevant tables are included here. The full set of generated outputs is included in the GitHub repository ZIP files.

V. Experiment 1 results: camera-only models

The camera-only experiment shows that adversarial training improved robustness substantially without hurting clean performance. The standard camera-only model achieved a best balanced validation accuracy of 94.73%, while the adversarially trained model reached 95.42%. Under clean conditions at epsilon = 0.05, both models performed similarly: 94.92% for the standard model and 94.07% for the adversarially trained model. Under attack the gap was big, the standard model dropped to 3.95% while the adversarially trained model held at 77.12%, a +73.16-percentage point improvement.

Model	Clean Accuracy	Attack Accuracy	Accuracy Drop
Standard Camera-Only	94.92	3.95	-90.97
Adv Trained Camera-Only	94.07	77.12	-16.95

Table 1. Camera-only model performance at epsilon = 0.05. Values are percentages.

The per-class breakdown shows a bigger picture. At epsilon = 0.05 the standard model nearly completely failed: 5.91% on straight and 0% on turn_left, turn_right, and stop. The adversarially trained model recovered all classes: 86.50% on straight, 29.27% on turn_left, 43.75% on turn_right, and 95.45% on stop. The turning classes remained the weakest, which makes sense given they are minority classes in the dataset, but the improvement was consistent across all four classes.

Class	N	Std Clean	Std Attack	Adv Clean	Adv Attack	Gain
straight	237	94.94	5.91	92.83	86.50	+80
turn_left	41	90.24	0.00	95.12	29.27	+29
turn_right	32	93.75	0.00	93.75	43.75	+43
Stop	44	100	0.00	100	95.45	+95

Table 2. Camera-only per-class accuracy at epsilon = 0.05. Values are percentages.

These results establish two important baselines. First, a well-trained camera-only model can fail almost completely under FGSM attack despite strong clean performance. Second, adversarial training fixes most of that vulnerability while keeping clean accuracy essentially the same. These numbers set the comparison point for the fusion experiment.

VI. Experiment 2 results: cross-attention fusion models

The fusion experiment compared a standard cross-attention fusion model against an adversarially trained cross-attention fusion model. The standard fusion model achieved a best-balanced validation accuracy of 93.40%, while the adversarially trained fusion model achieved 62.72%, much lower. Under attack at epsilon = 0.05, the standard fusion model completely collapsed to 0.28%, while the adversarially trained model held at 81.36%, a +81.07-percentage point improvement.

This result was surprising because prior papers on attention-based fusion often report strong performance under clean conditions. In this project, the standard fusion model also performed well under clean input, reaching 92.94% clean accuracy, although it was slightly below the standard camera-only model at 94.92%. However, under adversarial attack the fusion model performed worse than the camera-only model (0.28% vs 3.95%). One possible reason is that the corrupted camera query broke the cross-attention mechanism, making the model retrieve or emphasize the wrong LiDAR-sector information. The camera-only model at least had a direct and well-trained path from image features to predictions; the fusion model had a broken query trying to retrieve information from LiDAR tokens it could no longer match correctly.

Model	Clean Accuracy	Attack Accuracy	Accuracy Drop
Standard Fusion	92.94	0.28	-92.66
Adv Trained Fusion	81.92	81.36	-0.56

Table 3. Fusion model performance at epsilon = 0.05. Values are percentages.

The per-class results show that adversarial training helped most classes but not equally. The standard fusion model got 0% attack accuracy on all four classes. The adversarially trained model recovered strongly on straight (94%), stop (100%), and turn_left (51%), but turn_right remained very weak at only 3%. This class-specific weakness is the most notable limitation of the fusion model.

Class	N	Std Clean	Std Attack	Adv Clean	Adv Attack	Gain
straight	237	92	0	94	94	+93
turn_left	41	95	0	54	51	+51
turn_right	32	91	0	3	3.00	+3
stop	44	95	0	100	100	+100

TABLE 4. FUSION PER-CLASS ACCURACY AT EPSILON = 0.05. VALUES ARE PERCENTAGES.

The attention analysis adds an interesting layer to the results. In the standard fusion model, the L1 attention shift between clean and attacked conditions was 1.0215, and the top-sector change rate was 68.08% , meaning the model frequently changed which LiDAR sector it was attending to when the camera was attacked. This suggests the corrupted camera query was disrupting the entire fusion mechanism. In the adversarially trained model, the L1 attention shift dropped to 0.1166 and the top-sector change rate fell to 3.67%. Its attention pattern stayed much more consistent under attack, suggesting adversarial training helped stabilize how the model uses LiDAR even when the camera query is corrupted.

Model	L1 Attention Shift	Top-Sector Change Rate	Clean Entropy	Attack Entropy
Standard fusion	1.0215	68.08	1.4273	1.6604
Adv trained fusion	0.1166	3.67	1.2364	1.2987

TABLE 5. FUSION ATTENTION STABILITY UNDER CAMERA ATTACK. VALUES ARE PERCENTAGES.

VII. FINAL 2X2 COMPARISON

Table 6 summarizes the final 2×2 comparison at epsilon = 0.05. I focus on this epsilon because it was the value used during adversarial training and because it clearly showed the difference between standard and adversarially trained models. Adding LiDAR with standard training did not improve clean accuracy meaningfully, and adversarial training introduced a larger clean-performance cost for the fusion model than for the camera-only model.

Model	Clean Accuracy	Attack Accuracy	Accuracy Drop	Best Balanced Validation
Std Camera-Only	94.92	3.95	-90.97	94.73
Adv Camera-Only	94.07	77.12	-16.95	95.42
Std Fusion	92.94	0.28	-92.66	93.40
Adv Fusion	81.92	81.36	-0.56	62.72

TABLE 6. FINAL 2×2 CLEAN AND ATTACKED ACCURACY AT EPSILON = 0.05. VALUES ARE PERCENTAGES.

Overall, the final 2×2 comparison supports three main findings. First, clean accuracy alone is not a reliable indicator of adversarial robustness. Both standard models had high clean accuracy but collapsed under attack. Second, adversarial training was the dominant factor in improving robustness for both camera-only and fusion models. Third, fusion only provides an advantage when combined with adversarial training. At epsilon = 0.05, adversarially trained fusion outperformed adversarially trained camera-only by +4.24 percentage points, but at the cost of -12.15 percentage points lower clean accuracy.

This means the best model depends on the priority. If clean accuracy is the main goal, the adversarially trained camera-only model is stronger. If attacked accuracy under camera corruption is the main goal, the adversarially trained fusion model performs best overall. However, because of its weak turn_right performance, the fusion model would need further tuning before being considered consistently robust across all classes.

VIII. CONCLUSION

This project evaluated whether camera–LiDAR fusion and adversarial training improve robustness under camera-only FGSM attack. The standard camera-only and standard fusion models both performed well on clean data, but both failed under FGSM camera attack. This means clean accuracy alone did not predict adversarial robustness.

The strongest factor was adversarial training. It improved both camera-only and fusion models under attack, and the adversarially trained fusion model achieved the best overall attacked accuracy. However, fusion was not uniformly better: it had lower clean accuracy and performed poorly on turn_right, so the result should be interpreted as stronger overall robustness under attack, not perfect robustness.

There are also important limitations to consider when interpreting these numbers. The dataset used is small compared to what needs to be used or has been used in other papers. KITTI is a single city recorded in 2011, and the total training set had fewer than 1500 frames across four classes. The LiDAR branch in the fusion models was trained from scratch with no pretrained weights, unlike the camera branch which used ResNet18 pretrained on 14 million ImageNet images. This means the LiDAR branch was at a significant disadvantage throughout training and likely did not learn as discriminative a representation as it would with more data. With a larger and more diverse dataset like nuScenes, which has over 1.4 million frames from multiple cities, the LiDAR branch would have had a much better chance to learn strong 3D representations, which could have improved both the clean accuracy and the adversarial robustness of the fusion models.

It is also worth considering whether the numbers here would look different with more training time and stronger hardware. Both fusion models were trained for 15 epochs on Google Colab, which limits the batch size and number of training iterations possible. The adversarially trained fusion model reaching only 62.72% balanced validation accuracy suggests it may not have had enough training to fully converge. With more epochs and a larger dataset, the clean-robustness tradeoff might be less severe, and the turn_right class weakness might have been resolved. But I believe these limitations do not invalidate the main findings because the direction of the results is clear and consistent, but they do mean the specific numbers should be interpreted with caution.

The broader takeaway is that multi-sensor fusion should not be treated as a security feature on its own. It needs to be paired with adversarial training and evaluated specifically under adversarial conditions. For smaller AV companies that rely on LiDAR fusion to compensate for limited camera data, adversarial training on conflict scenarios is a practical and low-cost way to improve robustness without changing hardware or collecting more data if the training dataset is large enough to let both sensor branches actually learn.

IX. Code and Supplementary Material

The notebooks, generated outputs, prediction distributions, visual attack examples, and other files used in this project are available at:

https://github.com/Mayrinamelie/AV-Adversarial

X. Acknowledgments

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